



AQUALUMIN Nicheless Pool Light Replacement

Installation and Operations Manual



64-PAL-RAL	AquaLumin Sonar Retro Pool Light with Remote
64-PAL-RAL-80	AquaLumin Sonar Retro Pool Light with Remote and 80ft Cable
64-PAL-RAL-150	AquaLumin Sonar Retro Pool Light with Remote and 150ft Cable
64-PAL-RAL T	AquaLumin Sonar Retro Pool Light with Remote and PCR-2T-65 Transformer
64-PAL-RAL-80-T	AquaLumin Sonar Retro Pool Light with Remote, PCR-2T-65 and 80ft Cable
64-PAL-RAL-150	AquaLumin Sonar Retro Pool Light with Remote, PCR-2T-65 and 150ft Cable

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For your safety, it is imperative that this product be installed and maintained by a licensed and qualified contractor specializing in pool equipment, adhering to local regulations. The individual responsible for maintenance should possess extensive experience in pool equipment installation and upkeep to ensure precise adherence to the instructions outlined in this manual. Prior to installation, thorough comprehension, and compliance with all provided warning notices and instructions are crucial. Failure to do so may lead to property damage, personal injury, or loss of life. Any improper installation and/or operation may void the product's warranty and pose significant electrical hazards, potentially resulting in severe harm, property damage, or fatality. It is strictly advised not to modify this equipment. Installer, please be aware that this manual contains vital information concerning the installation, operation, and safe utilization of this product, which should be communicated to the equipment's owner/operator.

Section 1. SAFETY INSTRUCTION



Improper installation carries the potential for electrical hazards, presenting a risk of severe or even fatal injuries to pool users, installers, or bystanders due to the threat of electrical shock, and may also result in property damage.

Section 2. Preparation

This manual provides installation and operational instructions for the PAL Lighting AquaLumin Nicheless Replacement Pool Light.

Read the installation & operational manual before proceeding with the installation. Keep this manual for future reference.

Before starting the installation, check that you have the correct parts below. If any parts are missing, please call technical support at +1 (713)-946-4343.

2.1 Package Contents

AquaLumin Nicheless Replacement Pool Light

1 x Remote
1 x Remote Battery Pack
1 x Mounting Plate
1 x Sealing Plate
1 x Clear Dress Ring
1 x Waterproof Gel Pack
1 x Head Screw
3 x Small Mounting Screws
3 x Machine Screws

1 x Cable **Select Models*

1 x 24V Transformer **Select Models*

2.2 Preparation Work

1. The PAL AquaLumin Nicheless Replacement Pool Light is a 24V DC replacement light for the Pentair AquaLumin light. A 24V transformer will need to be installed to be able to operate our light.

2. Ensure that the pool's electrical system complies with the standards outlined in the current National Electrical Code (NEC).

IMPORTANT NOTICE

This PAL AquaLumin Nicheless Replacement Pool Light is intended to replace the existing AquaLumin Nicheless light. This product includes a new bracket, and a new light, completely replacing the original parts.

Section 3. Replacing the Light

A. Replacing the light and *optional cord*.

1. Remove the old AquaLumin from the niche. Undo the bolt at the top of the light. Take the gray plastic protector and all 4 bolts from the niche. See Figure 1.

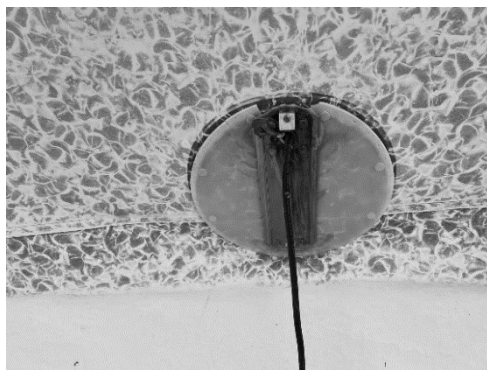


Figure 1

2. If the cable can be removed, remove the old cable, and leave a pull line through the conduit.
3. After removing all parts of the old AquaLumin, place the gray gasket plate closest to the liner. The light mount with the words TOP should face outward.

4. Align the screws with the niche. See Figure 2 and 3.

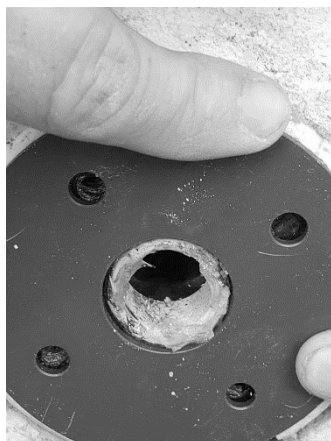


Figure 2



Figure 3

NOTE: The three machined screws will attach to the light mount and go through the gray plate into the niche to secure the light mount. The fourth pan head screw will secure only the gray gasket plate.

NOTE: The niche may cause the light mount to not align straight up and down. To fix this, turn the light mount to align with the screw hole and attach.

5. Once the gray face plate and light mount have been attached, pull the attached cable through the light mount and conduit back to a 24v transformer.

NOTE: When wiring into the transformer, there will be a brown and blue terminal for the wires to screw into on the left side.

6. Attach the light to the mounting plate with the thin screws at the 1, 3, and 7 o' clock positions. See Figure 4.

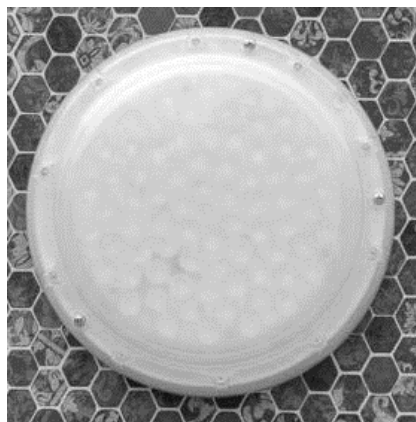


Figure 4

IMPORTANT NOTICE

AquaLumin Nicheless Replacement Pool Light. However, you can use the PAL gel kit and the existing wires to wire into the AquaLumin Nicheless Replacement Pool Light.

B. Installing Cable on AquaLumin with Gel Pack.

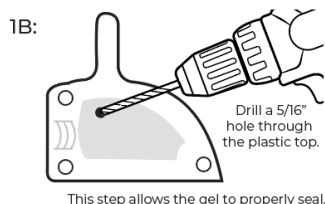
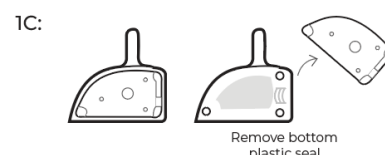
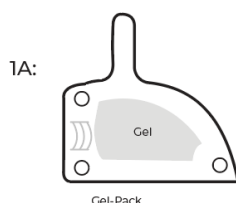
Note: The gel pack can only be installed once. Once the screws are inserted, they must not be removed or reset, as this will compromise the water-tight seal.

1: Prepping the gel pack.

- Inspect gel pack to ensure no visible cracks/chips. (1A)

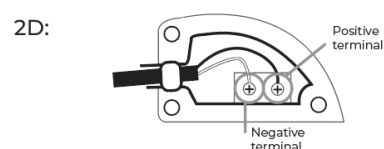
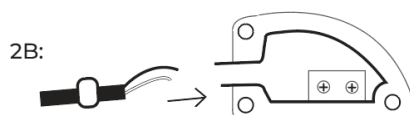
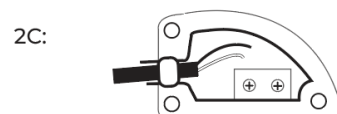
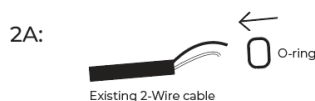
★ **For the gel pack to seal properly, drill a 5/16" hole into the top face of the gel pack.** The hole will need to avoid all sidewalls of the gel pack. The hole allows space for the gel to push excess air out and create a water tight seal. (1B) See illustration for approx. hole location.

- Ensure the hole goes all the way through.
- Remove the bottom plastic seal and set gel pack aside. the Plastic seal can be discarded. (1C)



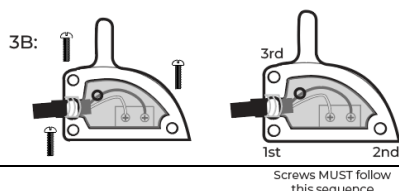
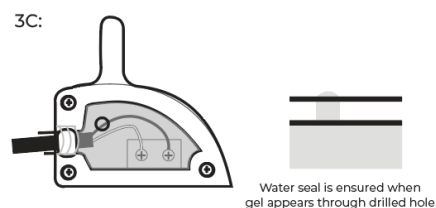
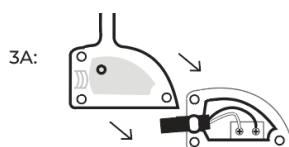
2: Wiring

- Cut the cable going to the existing light fixture.
- Push included O-ring through cable. (2A)
- Expose 2-Wire cables to install to the terminals on the back of the PAL light fixture.
- Wires should be stripped back enough to make connection in the terminal block, but not so much that they are exposed when exiting the gel pack. (2B and 2C)
- Place the positive cable into the furthest screw from the cable entry. Place the negative cable into the closest screw from the cable entry. (2D).



3: Attaching the gel pack.

- Line up the gel pack with the terminal block on the back of the light fixture. (3A)
- ★ The gel pack will fit very snug, install the screws to attach the gel pack to the light fixture terminal block. **The screws MUST be installed following the sequence below.** (3B)
- The gel pack is fully installed when the O-ring is set properly in the cable feed channel and the drilled hole has gel pushing through. (3C)



Section 4. REMOTE PROGRAMMING



Pairing Lights to Zones:

Within 3 seconds of turning light on, press desired channel number's  button 3 times. Once linked, light(s) will flash 3 times.

Cloning Modes:

Within 3 seconds of turning light on, press and hold  button on required channel:

Channel 1 - **PAL**[™]

Channel 2 - **Jandy**[™]

Channel 3 - **HAYWARD**[™]

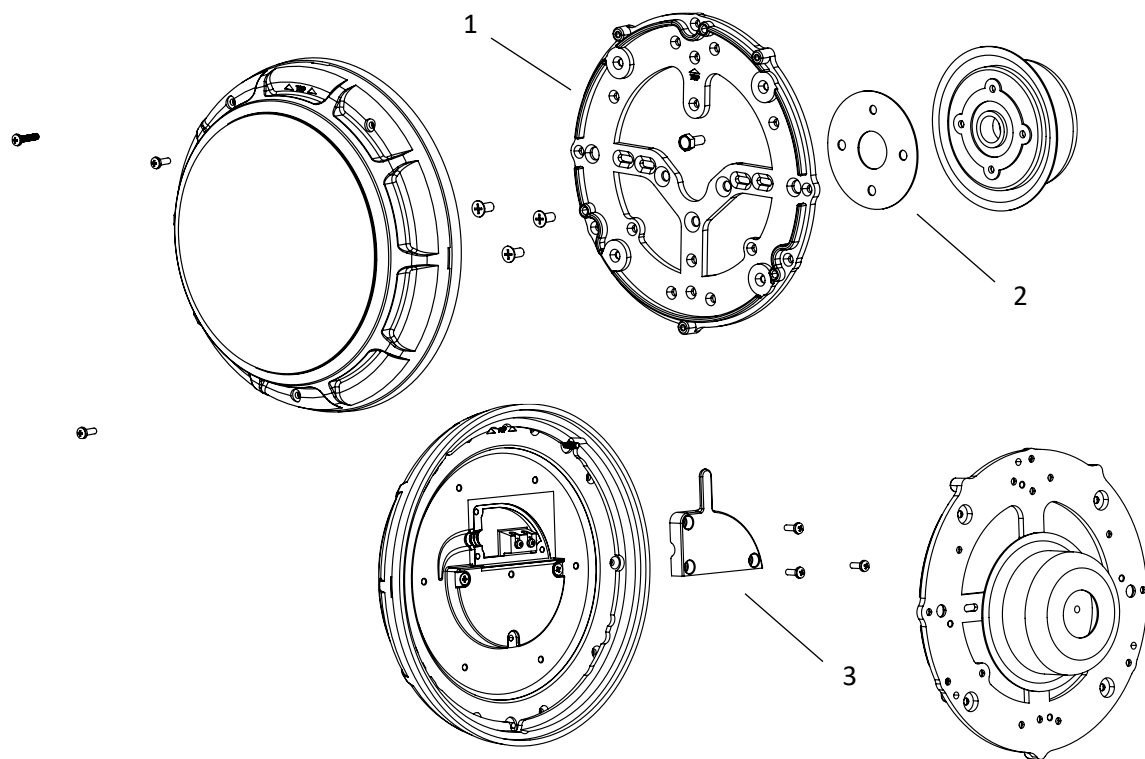
Channel 4 -  **PENTAIR**

Unpairing Lights from Zones:

Within 3 seconds of turning light on, press the channel zone number's  button that the light is currently linked to 5 times. Light will flash 9 times quickly to confirm it has been unlinked.

Section 5. REPLACEMENT PARTS

Light Replacement Parts



Replacement Parts		
Item	Part Number	Description
1	64-PAL-RAL-BRK	Wall Mounting Bracket
2	64-PAL-RAL-GASKET	Wall Spacer
3	64-PAL-RAL-GEL	Gel Pack
	64-PAL-RAL-SCREW	Screw Bag

[illegible]

